Writing and documenting code

VS Code

Jupyter

PyCharm

Positron

Spyder

Org Mode (if you enjoy suffering)

A brief history

Pre-2000s

— command line, Perl, text editors like `vi` and `emacs`

2000s

— R rises, RStudio (2011 - real IDE); MATLAB popular in quantitative fields

2010s

— Python overtakes Perl; Jupyter Notebook (2014) used for sharing analyses

Now

— Python and R displaced Perl, other languages and development tools available

What is Jupyter?

- Stands for **Ju**lia, **Py**thon, **R** — designed to support all three from the start
- A **server-client application** — your browser is the interface, a kernel runs the code
- The document format (`.ipynb`) mixes **code cells, markdown text, and output** in one file
- Currently among the most popular development environments used by data scientists due to its flexibility
- Two flavors: **Jupyter Notebook** (classic) and **JupyterLab** (modern, more IDE-like)

Pros and Cons

Interactive

Narrative: thoughts and code together

Reproducible

Inline visualization: plots by their code

Low barrier: easy to start

Hidden state problem: cells can be run out of order

Version control is painful: `.ipynb` files are JSON under the hood, so `git diff` is nearly unreadable

Not built for software: writing reusable modules, packages, or pipelines is messy

Debugging is limited

Beginners learn bad habits: code that only works in one specific order

The hidden state problem, illustrated

```
# Cell 1 — run this first
x = 10

# Cell 2 — run this second
x = x * 2

# Cell 3 — now delete Cell 1 and re-run Cell 2
# x is still 20 in memory, so the answer is wrong
# restart the kernel and everything falls apart
```

This is the most common source of "it worked yesterday" problems in Jupyter. **Restart and run all** before you trust your own notebook.

Other options

Tool	Best for
VS Code + Jupyter	General purpose, notebooks + scripts, strong debugging
Positron	RStudio expats; built by the Posit team; plot pane, variable explorer
Spyder	Simple, scientific, very RStudio-like; popular via Anaconda
PyCharm	Software engineering; heavy but powerful
Google Colab	Zero install; GPU access; great for sharing and teaching
Org Mode	Hard Mode

VS Code + Jupyter: a popular option

You get:

✅ Inline plots and cell-by-cell execution (like classic Jupyter)
✅ Proper file and project management
✅ Real debugging tools
✅ Version control that actually works
✅ The same editor you'll use for `.py` scripts, pipelines, and everything else

❌ additional setup
❌ less intuitive interface
❌ still has the hidden state caution

We will not cover this, but instructions are at the end if you want to use it

Jupyter on Nova

Open your terminal and SSH to Nova

```
ssh <netid>@nova.its.iastate.edu
# Enter your 2FA code and then password

# Request an interactive session:
salloc -N 1 -n 4 -t 2:00:00 -p interactive
```

You requested a single node (computer) with 4 processors (CPUs) for 2 hours

Load the module for conda

```
module load micromamba
conda create -n notebook

# proceed with "Y" when prompted
eval "$(micromamba shell hook --shell=bash)"
micromamba activate notebook
(notebook) conda install -c conda-forge jupyterlab ipython \
                                ipykernel notebook
# proceed with "Y" when prompted
```

Success

```
Transaction finished
To activate this environment, use:
    $ [micro]mamba activate <environment>
Or to execute a single command in this environment, use:
    $ [micro]mamba run -n <environment> mycommand
```

Active your notebook (still interactive)

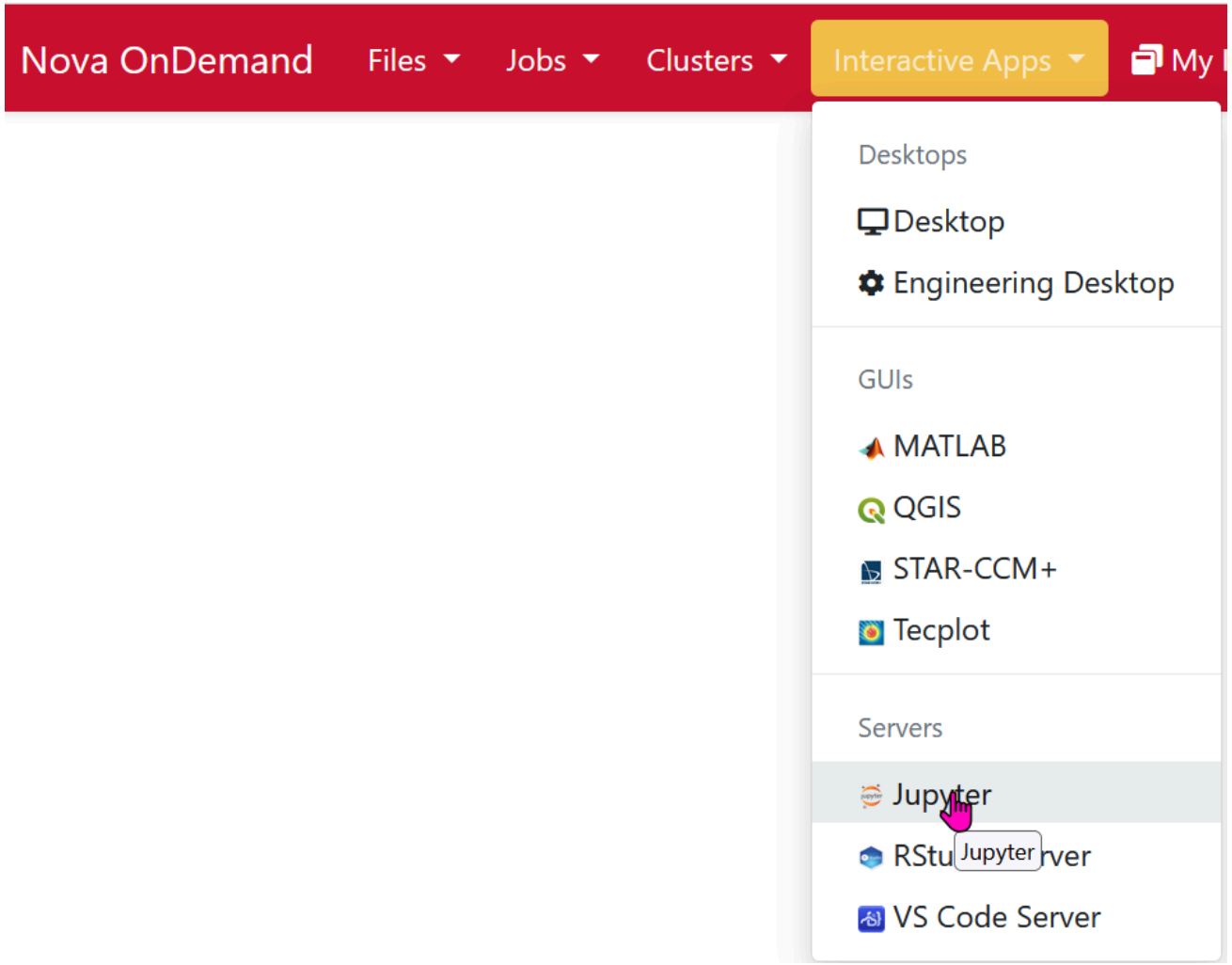
```
micromamba activate notebook

# Add the env to jupyter kernels
python -m ipykernel install --user --name notebook --display-name "Python3 (EEOB546)"

# Next, install some useful packages
conda install -c conda-forge pandas statsmodels numpy \
                matplotlib seaborn scipy lxml biopython

# when done
exit
```

Enter your notebook at https://nova-ondemand.its.iastate.edu

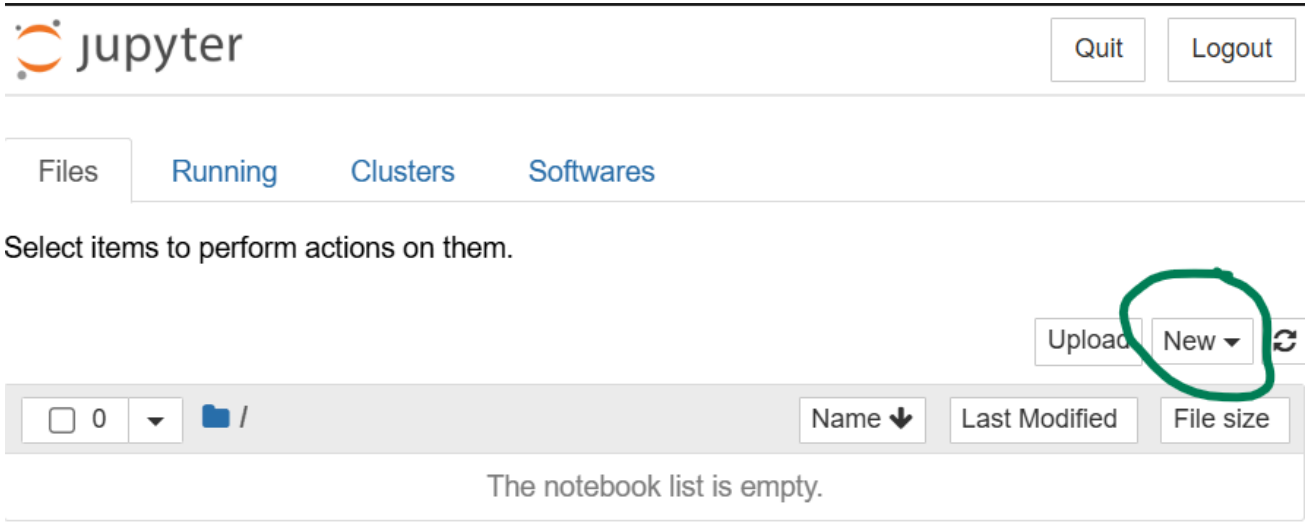


Fill out the form

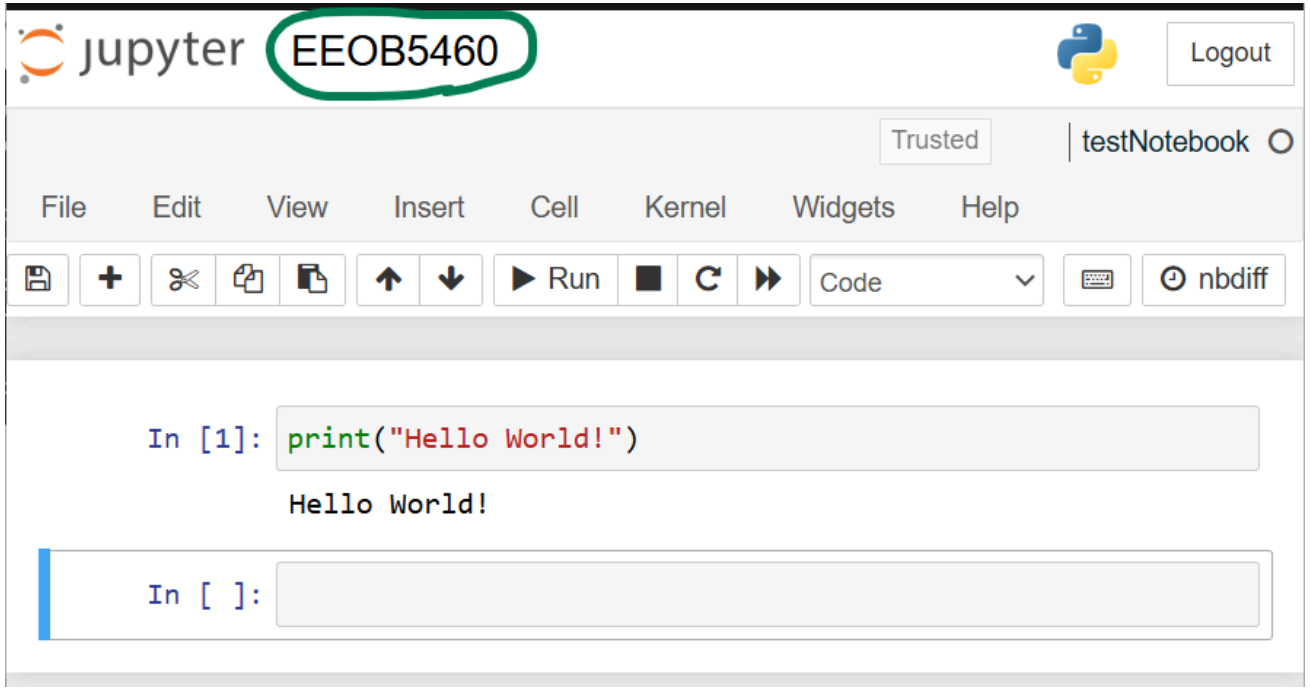
```
Account: <s2026.eeob.546.1>
Queue: <instruction>
Number of hours: <2>
Memory required: <32G>
Number of tasks per node: <4>
GRES (only applies to GPU jobs): <blank>
Job name: <jupyter>
<check>: I would like to receive an email
Jupyter Notebook vs Lab: <Notebook>
Working Directory: <wherever you want>
```

Then launch the session by clicking Launch.

Jupyter


jupyter
Quit
Logout
Files
Running
Clusters
Softwares
Select items to perform actions on them.
Upload
New
0
/
Name
Last Modified
File size
The notebook list is empty.

Rename and test your notebook



Jupyter Notebook Tips

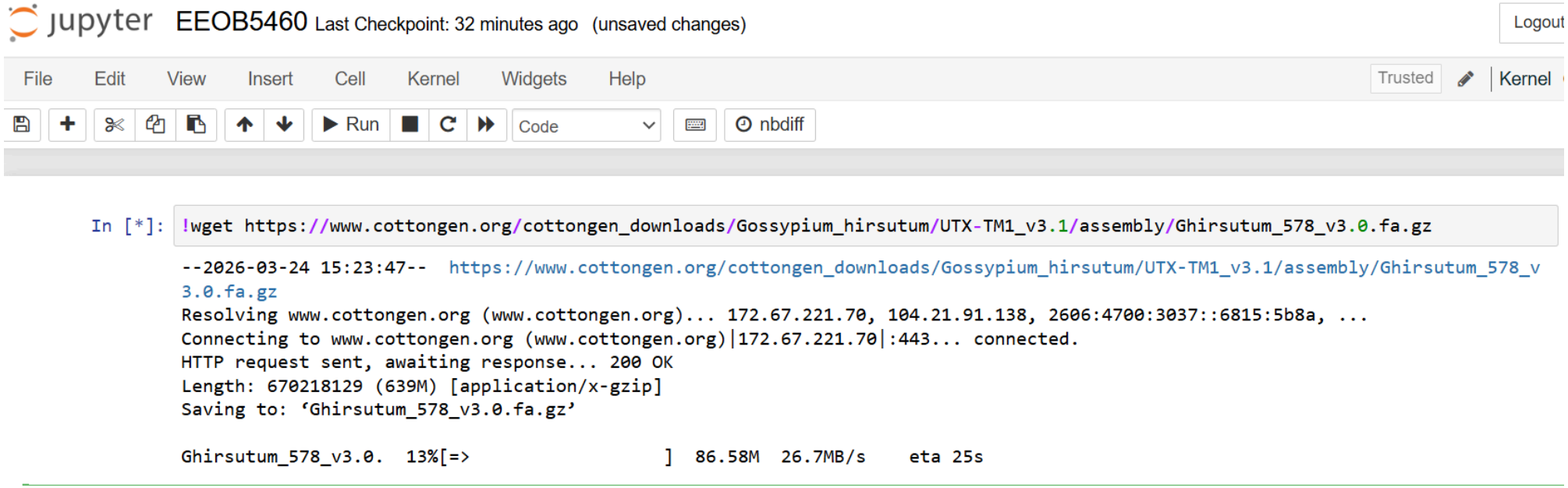
1. Tab is your friend
2. Use `!` to call bash from the notebook
3. Magic commands exist
4. Master the keyboard shortcuts
5. Help is closer than you think
6. Other tips

Use Tab (here and on commandline)

- Tab completion is available in Jupyter Notebook
  -- Complete variable names, functions, and methods
  -- Explore the structure of objects
  -- Discover methods and attributes of objects

! can run bash commands within Jupyter

- Note: not all commands work, but the important ones do



Maybe useful magic commands

Command	Operation
`%lsmagic`	List all magic commands
`%time`	Measure execution time
`%history`	View command history
`%env`	Set environment variables
`%memit`	Check memory usage
`%reset -f`	Clear all variables
`%debug`	Start the debugger
`%run script.py`	Run an external Python script

Master the Keyboard Shortcuts

Run cells

- Ctrl + Enter → run, stay in cell
- Shift + Enter → run, next cell
- Alt + Enter → run, new cell below

Add / manage cells

- Esc + A / B → add above / below
- Esc + D, D → delete cell

Cell types

- M → Markdown
- Y → Code

Editing

- Ctrl + / → comment / uncomment

Tip

- Press **H** to view all shortcuts
- You can customize shortcuts

Getting Help

- Add `?` or `??` after a function to view documentation
- Press **Shift + Tab (twice)** to see docstring description
- Use `help(function)` for a detailed description
- Use `dir(object)` to list available attributes/methods

Other Jupyter Notebook features

- **Markdown**
  Format text, add images, links, and equations for clear, readable notes
- **Exporting**
  Save notebooks as HTML, PDF, slides, or a `.py` script
- **Widgets**
  Add interactive elements like sliders, buttons, and dropdowns
- **Extensions**
  Install add-ons to expand functionality (e.g., variable inspector, themes)

That's it for this lecture, but keep reading if you want to use VS Code, with or without Jupyter Notebook

If you are interested in setting up VS Code and Jupyter

Note: since I am not teaching this, I asked Claude (AI) to generate these slides. I've reviewed them, but some of the Mac stuff may not work (I use Windows)

Setting Up VS Code with Jupyter

Everything you need to write and run Python with inline visualizations

Step 1: Install Python

All platforms: Download from python.org — get the latest stable version

⚠️ **Windows only:** During install, check **"Add Python to PATH"**

Verify it worked — open a terminal and run:

```
python --version
# Python 3.12.0
```

A Note on Terminals

You'll be using a terminal throughout this course

OS	App to open
Windows	Command Prompt or PowerShell
Mac	Terminal
Linux	Terminal

VS Code has a built-in terminal

Step 2: Install VS Code

- Download from code.visualstudio.com
- Default install options are fine on all platforms

Step 3: Install the Python Extension

1. Open VS Code
2. Click the **Extensions** icon in the left sidebar (or `Ctrl/Cmd + Shift + X`)
3. Search **"Python"**
4. Install the one by **Microsoft**

This also installs **Pylance** automatically — that's expected and good

Step 4: Install the Jupyter Extension

Same process:

1. Extensions panel
2. Search **"Jupyter"**
3. Install the one by **Microsoft**

This is what gives us inline plots and interactive cells

Step 5: Install the Required Packages

Open the VS Code terminal and run:

```
pip install jupyter notebook matplotlib numpy pandas
```

If `pip` doesn't work on Windows, try:

```
python -m pip install jupyter notebook matplotlib numpy pandas
```

This will take a minute — that's normal

Step 6: Select Your Python Interpreter

VS Code needs to know which Python to use

1. Press `Ctrl/Cmd + Shift + P`
2. Type **"Python: Select Interpreter"**
3. Choose the version you installed from python.org

If you see multiple options and aren't sure — pick the one that shows the version number you installed

Step 7: Create Your First Notebook

1. Open a folder in VS Code (`File → Open Folder`) — always work inside a folder
2. Create a new file and name it `test.ipynb`
3. VS Code will open it as a notebook automatically

Click "**+ Code**" to add a cell and type:

```
print("it works")
```

Click the ▷ button or press `Shift + Enter` to run it

Step 8: Test Inline Plotting

Add a new cell and run this:

```
import matplotlib.pyplot as plt

plt.plot([1, 2, 3, 4], [1, 4, 9, 16])
plt.title("My first plot")
plt.show()
```

The plot should appear **directly below the cell**

If it does — you're all set

The Two Things to Remember

Always open a folder, not a file
`File → Open Folder` keeps your work organized and prevents a lot of headaches

Run cells with `Shift + Enter`
It runs the current cell and moves to the next one — you'll use this constantly

If Something Went Wrong

Most setup problems fall into one of these:

- **Python not found** → Windows PATH issue, re-run the installer and check the box
- **pip not found** → use `python -m pip` instead
- **Plots not showing inline** → make sure the Jupyter extension is installed and you're in a `.ipynb` file, not a `.py` file
- **Wrong interpreter selected** → `Ctrl/Cmd + Shift + P` → "Python: Select Interpreter"